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Short communication

Student ERI: Psychometric properties of a new brief measure of effort-reward imbalance among university students

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ABSTRACT

Background: Psychosocial stress among university students, particularly medical students, is considered a widely prevalent problem. There is a need for valid measurement of an adverse psychosocial stress environment in university settings. The aim of this study was to examine the psychometric properties of a newly developed short student version of the effort-reward imbalance (ERI) questionnaire in a sample of medical students.
Methods: A cross-sectional survey with a self-administrated questionnaire containing three scales was conducted among 406 medical students. Item-total correlations and Cronbach's alpha were calculated to assess the internal consistency of the scales. Confirmatory factor analysis was applied to test factorial validity of the questionnaire structure.
Results: The student version of the ERI questionnaire provides acceptable psychometric properties. The Cronbach's alpha coefficients for effort, reward, and over-commitment were 0.67, 0.65, and 0.79, respectively. Confirmatory factor analysis displayed a satisfactory fit of the data structure with the theoretical concept (GFI > 0.94).
Conclusions: This student version of the ERI questionnaire provides a psychometrically tested tool for studies focussing on psychosocial environment in university settings. Further applications of this approach in other student groups are needed, in addition to prospective studies assessing associations with health outcomes.

1. Introduction

Stressful psychosocial environments are associated with a number of risks. Previous studies suggest that a stressful environment in university settings may affect health, academic performance, and further career development in university students [1–6]. Providing a theory-based measure of a stressful psychosocial environment in university settings is important in order to assess related risks and develop health-promoting settings.

Several theoretical models of stressful psychosocial work environments have been developed and applied in occupational health research [7–12]. One of the most widely tested models, 'effort-reward imbalance' (ERI), posits that the imbalance between high effort spent and low reward received in turn elicits strong negative emotions and sustained stress responses [11–13]. These effects may be amplified in the presence of a distinct personal coping characteristic, termed over-commitment. A robust body of epidemiological evidence obtained from employed populations indicates that effort-reward imbalance is associated with elevated risks of poor physical and mental health functioning [14–20].

Recent studies have extended this work-related model to the context of school work among adolescents by adapting the original ERI questionnaire to the school setting and revealed sound psychometric properties [17,22]. Furthermore, the studies indicate significant adverse health effects of perceived effort-reward imbalance in school-aged children and adolescents, such as depressive symptoms [22], suicidal ideation [23], fatigue [24], somatic pain [21], and poor self-rated health [17,21].

Psychological distress among university students is widely prevalent, given highly challenging curricula, an intensive and time-consuming workload, high intellectual and emotional demands, and lack of leisure time and recreation, particularly among medical students. Identifying stressful components within this complex psychosocial environment is crucial not only in scientific terms, but also in terms of developing preventive strategies. There is a need for reliable and valid measures of health-adverse psychosocial conditions in university settings. The intention of this study is to provide a reliable short theory-based measure of a stressful psychosocial environment in university settings, with items reflecting all dimensions of the approved original

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effort—reward imbalance model. The aim of this investigation was to examine psychometric properties of a newly developed standardized self-administered short version of the ERI questionnaire, using a sample of medical students. In this short report we focus mainly on the criteria of content and construct validity.

2. Methods

2.1. Study design and participants

This current investigation is based on a cross-sectional survey which is a part of the prospective cohort study "Healthy Learning in Duesseldorf" (HeLD). HeLD-Study was initiated at the start of a major reform of the curriculum of medical education at the Medical Faculty of Heinrich Heine University Duesseldorf in Germany. All medical students, enrolled at the Heinrich-Heine-University of Dusseldorf during 2012–2013, were invited to participate. In total, 406 students completed our questionnaire survey during the second semester of the first year, with a response rate of 50.9%.

2.2. Measurements

Information was obtained through a self-administered, newly developed short version of the ERI questionnaire. The items on this questionnaire were designed in German language in a way to preserve the original meaning while capturing the specific characteristics of an adverse psychosocial environment in university settings [13,17]. In compliance with the authors of the original ERI short version each item was evaluated with regard to its ability to mirror the distinct work environment of learning and training in a University setting. A majority of items could be easily applied to this setting. However, some items had to be modified as they did not match with this context, as obvious from qualitative interviews that we conducted with focus groups. For instance, items concerning monetary reward did not matter in this context, whereas rewards related to esteem and favourable social comparison were relevant. Thus, we designed a new respective item '*rew2*'. Table 1 displays the exact wording of the items.

The scale 'effort' is measured by 3 items, the scale 'reward' by 6 items, and the scale 'over-commitment' by 5 items using a 4-point Likert scaled answer format ('strongly disagree' [1], 'disagree' [2], 'agree' [3], 'strongly agree' [4]). Average scores of these ratings were calculated with appropriate recoding, and an effort-reward ratio was computed by dividing the 'effort' score by the 'reward' score, using the established algorithm [13]. High values reflect high effort, high reward, high over-commitment, and high effort-reward ratio, respectively.

The study was approved by the University of Duesseldorf institutional ethic committee. The authors assert that all procedures contributing to this work comply with the ethical standards of the relevant national and institutional committees on human experimentation and with the Helsinki Declaration of 1975, as revised in 2008. All participants included in the survey submitted written informed consents. Students were informed about the study and their rights as potential participants.

2.3. Statistical analysis

Established procedures for testing the psychometric properties of the scales were used. Firstly, items and scale means were computed. Item-total correlations and Cronbach's alpha were calculated to assess the internal consistency of the scales. Confirmatory factor analysis was performed to test the dimensional structure of the theoretical model, and goodness-of-fit indices were calculated for the scales. We used the goodness-of-fit index (GFI), which indicates the amount of variance and covariance explained by the model (values over 0.90 indicating an acceptable model fit), the adjusted-goodness-of-fit index (AGFI, which adjusts the GFI for degrees of freedom in the model, with values above 0.90 indicating a good fit to the data), and the Root Mean Square Error of Approximation (RMSEA) criterion. The comparative fit index (CFI) was used as a measure of comparative fit, with values exceeding 0.90 indicating a good fit to the data. IBM-SPSS 22 for the statistical analyses was used.

3. Results

The mean age of the initial sample of students was 21.1 years (SD = 3.9), 30% were men. 34.4% of participants had both parents without an academic degree while the majority of students came from families with academic background. Mean scores of the scales effort, reward and over-commitment, and the mean value of the effort-reward ratio were 3.09 (SD = 0.51), 2.94 (SD = 0.44), 2.63 (SD = 0.68) and 1.09 (SD = 0.32), respectively.

3.1. Internal consistency

Item-total correlation for each scale, together with scale Cronbach's alpha are presented in Table 1. The Cronbach's alpha coefficients for effort, reward, and over-commitment were 0.67, 0.65, and 0.79, respectively. Item-total correlations varied between 0.30 and 0.73.

Table 1
Mean, SD, item-total correlation and Cronbach's alpha coefficients of the ERI scales.

Items	Mean (SD)	Item-total correlations	Alpha of scale
Effort (eff1-eff3)	3.09 (0.51)		0.67
I have constant time pressure due to a heavy study load (eff1).	3.61 (0.59)	0.56	
I have many interruptions and disturbances while preparing for my exams (eff2).	2.42 (0.72)	0.41	
My study load has become more and more demanding (eff3).	3.19 (0.81)	0.50	
Reward (rew1-rew6)	2.94 (0.47)		0.65
I receive the respect I deserve from my supervisors (teachers) (rew1).	2.36 (0.84)	0.42	
I receive the respect I deserve from my fellow students (rew2).	3.01 (0.65)	0.38	
I am treated unfairly at university (rew3).[a]	2.84 (0.92)	0.35	
I am not sure whether I can successfully accomplish my university training s (rew4).[a]	3.04 (0.87)	0.35	
Considering all my efforts, I receive the appreciation that I deserve (rew5).	2.91 (0.66)	0.50	
Considering all my efforts and achievements, my job promotion prospects are adequate (rew6).	3.36 (0.65)	0.30	
Over-commitment (oc1-oc5)	2.63 (0.68)		0.79
As soon as I get up in the morning I start thinking about study problems (oc1).	2.82 (0.95)	0.64	
When I get home, I can easily relax and "switch off" from study (oc2).[a]	2.61 (0.93)	0.51	
People close to me say I sacrifice too much for my study (oc3).	2.41 (0.92)	0.49	
Student work rarely lets me go; it is still on my mind when I go to bed (oc4).	2.97 (0.86)	0.73	
If I postpone something that I was supposed to be done today I'll have trouble sleeping at night (oc5).	2.34 (0.94)	0.52	
Effort-reward ratio	1.09 (0.36)		

Table displays the exact wording of the items. Answer format—4-point Likert scale: [1] 'strongly disagree', [2] 'disagree', [3] 'agree', [4] 'strongly agree'.
[a] Reversed items (rew3, rew4, oc2): [1] 'strongly agree', [2] 'agree', [3] 'disagree', [4] 'strongly disagree'.

3.2. Factorial validity

The factorial validity of the ERI scales was tested with confirmatory factor analysis. Fig. 1 shows the summary of the confirmatory factor analysis—the model reached an acceptable level of fit (GFI > 0.90, AFGI > 0.90, RMSEA < 0.08, CFI > 0.90).

4. Discussion

The current findings show that the ERI model, which was initially designed to identify a health-adverse psychosocial environment of paid work, can be effectively adapted to a different working context, as represented by an intense learning and training environment within Universities, and specifically within a Medical Faculty. Using a 14-item questionnaire, we replicated the 3 core theoretical components of the model (effort, reward and over-commitment), where confirmatory factor analysis documented a satisfactory fit of the data structure with the theoretical concept.

The Cronbach's alpha coefficients for the three scales of effort, reward and over-commitment in our study are somewhat lower compared with working population from previous reports [13,16,17]. In part, this deviation might be due to the small number of items and, in case of 'reward' to the fact that this scale is composed by 3 sub-scales representing distinct components of reward [25,26]. The factorial structure of the scales of the ERI questionnaire of our study is in line with the one identified in previous psychometric reports of the ERI questionnaire [13]. A good fit of the data (Fig. 1) with the theoretical structure of the ERI model supports to some extent the structural validity of this measurement approach. To our knowledge, this is the first report to measure an adverse psychosocial university environment in terms of ERI among medical students.

Several limitations should be mentioned. First, the results are restricted to one group of medical students of the selected University in one country, Germany. It is not known whether they can be generalized to students in other faculties or in other university settings. Moreover, we were not in a position to analyse potential selection bias within the sample of this study, a sample that was remarkably homogeneous in socioeconomic and socio-cultural terms. Second, this information is based on a single measurement wave, and no extended test of psychometric properties could be performed, such as test-retest reliability or criterion validity, using prospective information on health outcomes. Third, the factor loading of some items (e.g. item '*rew6*') was critically low. When we replicated CFA by removing item '*rew6*' a modest improvement of fit indices only had been achieved (results not shown). Clearly, this leaves some room for improved item operationalization. Further research is also recommended concerning concurrent validity of this measurement approach, a further limitation of this study. For instance, the complementary work stress model of demand-control has been analysed in parallel to the effort-reward imbalance model in school children in Sweden [21]. We also included a proxy measure of the complementary work stress model of demand-control in our study protocol, but the correlation of this measure with effort-reward imbalance was relatively weak ($r = 0.332^{**}$). Given clear differences between the two models at the conceptual and operational level, we did not expect to test concurrent validity by this approach [11,27]. Although we could not offer data on criterion validity in terms of health previous research on psychosocial stress at school in adolescents and its association with health demonstrated very similar results between the demand-control model and the effort-reward imbalance model [17,21–24]. Despite these limitations, we were able to demonstrate satisfactory psychometric properties of a short, theory-based self-report assessment of a stressful working environment of University students. Given a high prevalence of psychological distress among University students improved monitoring of potentially harmful psychosocial environments is recommended as a basis of targeted intervention in vulnerable groups.

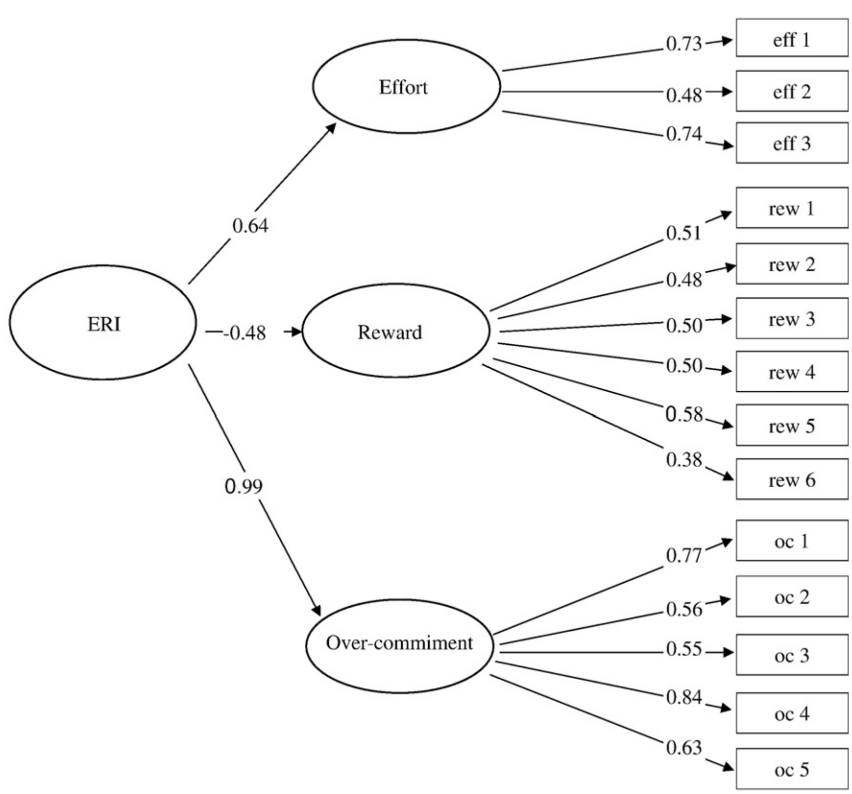

Fig. 1. Confirmatory factor analysis testing the underlying theoretical ERI construct. (ERI—effort-reward imbalance; eff1-eff3—effort items 1–3; rew1-rew6—reward items 1–6; oc1-oc5—over-commitment items 1–5).

Conflicts of interest

None declared.

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